

Homework 3 Solutions

Based on Lecture 3: Exponential Functions.

Exercise 1 (routine). Simplify each using the addition and quotient rules:

- $\frac{2^5 \cdot 2^{-7}}{3^{x+2}}$
- $\frac{3^{x-1}}{5^{2x} \cdot 5^{-3}}$
- $\frac{5^{2x} \cdot 5^{-3}}{5^{x-4}}$

Using $b^m b^n = b^{m+n}$ and $b^m / b^n = b^{m-n}$:

- $2^5 \cdot 2^{-7} = 2^{5-7} = \boxed{2^{-2} = \frac{1}{4}}.$
- $\frac{3^{x+2}}{3^{x-1}} = 3^{(x+2)-(x-1)} = 3^3 = \boxed{27}.$
- $\frac{5^{2x} \cdot 5^{-3}}{5^{x-4}} = 5^{2x-3-(x-4)} = 5^{x+1} = \boxed{5^{x+1}}.$

Exercise 2 (algebraic). Simplify each completely:

- $(2^x)^3 \cdot 2^{-2x}$
- $(3^x)^2 \cdot 3^{-x+1}$
- $\frac{(2^{2x})^{-1} \cdot 2^{x+5}}{4^{x-1}}$
- $(2^x)^3 \cdot 2^{-2x} = 2^{3x} 2^{-2x} = \boxed{2^x}.$
- $(3^x)^2 \cdot 3^{-x+1} = 3^{2x-x+1} = \boxed{3^{x+1}}.$
- Rewrite $4^{x-1} = 2^{2x-2}$. Then

$$\frac{(2^{2x})^{-1} \cdot 2^{x+5}}{4^{x-1}} = \frac{2^{-2x} 2^{x+5}}{2^{2x-2}} = 2^{-2x+x+5-(2x-2)} = \boxed{2^{7-3x}}.$$

Exercise 3 (equations). Solve for x in each. (Use that $b^x = b^y$ implies $x = y$ when $b > 0$, $b \neq 1$.)

- $5^{2x-1} = 5^{x+4}$
- $2^{3x} = 2^{x+8}$
- $4^{x+1} = 8^{2x-1}$ (write both sides as powers of 2 first)
- $2x - 1 = x + 4$, so $\boxed{x = 5}.$
- $3x = x + 8$, so $2x = 8$ and $\boxed{x = 4}.$
- Write both sides with base 2:

$$4^{x+1} = 2^{2x+2}, \quad 8^{2x-1} = 2^{6x-3}.$$

Thus $2x + 2 = 6x - 3$, so $5 = 4x$ and $\boxed{x = \frac{5}{4}}.$

Exercise 4 (rational exponents). Evaluate each exactly, then rewrite any negative exponent as a reciprocal:

- $16^{3/4}$
- $27^{-2/3}$
- $81^{1/2}$

- $16^{3/4} = (\sqrt[4]{16})^3 = 2^3 = \boxed{8}$.
- $27^{-2/3} = 1/(27^{2/3}) = 1/(\sqrt[3]{27})^2 = \boxed{\frac{1}{9}}$.
- $81^{1/2} = \sqrt{81} = \boxed{9}$.

Exercise 5 (features). State the domain, range, y -intercept, and horizontal asymptote of $y = 5^x$, and say whether it is increasing or decreasing.

The base 5 is positive and greater than 1, so 5^x is defined and positive for every real x , and is increasing. Also $5^0 = 1$, and as $x \rightarrow -\infty$, $5^x \rightarrow 0^+$ without reaching zero. Thus the domain is $\boxed{(-\infty, \infty)}$, the range is $\boxed{(0, \infty)}$, the y -intercept is $\boxed{(0, 1)}$, the horizontal asymptote is $\boxed{y = 0}$, and the function is $\boxed{\text{increasing}}$.

Exercise 6 (comparison). Without a calculator, decide which is larger: $3^{0.4}$ or $3^{0.5}$? Which is larger: $(1/3)^{0.4}$ or $(1/3)^{0.5}$? Explain using monotonicity.

Because $3 > 1$, 3^x is increasing, and $0.5 > 0.4$. Hence $\boxed{3^{0.5} > 3^{0.4}}$. Because $0 < 1/3 < 1$, $(1/3)^x$ is decreasing, so $\boxed{(1/3)^{0.4} > (1/3)^{0.5}}$.

Exercise 7 (sketch). Sketch $y = (1/3)^x$ on the same axes as $y = 3^x$. Label their common point, the horizontal asymptote, and the points with $x = 1$ and $x = -1$ on each graph.

Since $(1/3)^x = 3^{-x}$, this graph reflects $y = 3^x$ across the y -axis. Both pass through $(0, 1)$ and have horizontal asymptote $y = 0$. The requested points are $(1, 3)$ and $(-1, 1/3)$ on $y = 3^x$, and $(1, 1/3)$ and $(-1, 3)$ on $y = (1/3)^x$.

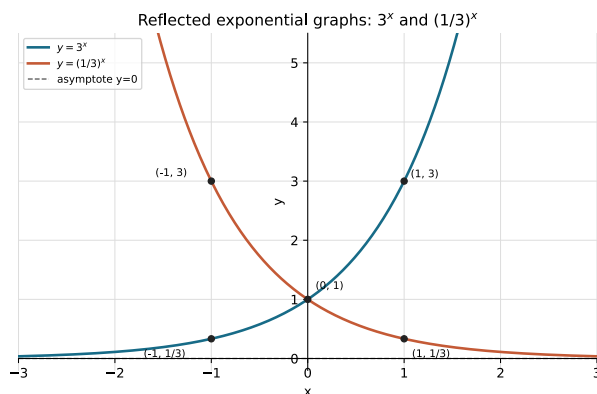


Figure 1: Graphs of $y=3^x$ and $y=(1/3)^x$, showing their reflection symmetry, shared point $(0,1)$, points at $x=-1$ and $x=1$, and horizontal asymptote $y=0$

Exercise 8 (routine). \$500 is invested at 4% annual interest, compounded continuously. Write the formula for $A(t)$ and find the balance after 20 years (leave the answer in terms of e , then approximate using $e^{0.8} \approx 2.2255$).

For continuous compounding, $A(t) = Pe^{rt}$. Here $P = 500$ and $r = 0.04$, so

$$A(t) = \boxed{500e^{0.04t}}.$$

After 20 years,

$$A(20) = 500e^{0.04(20)} = \boxed{500e^{0.8}} \approx 500(2.2255) = \boxed{\$1112.75}.$$

Exercise 9 (calculator comparison). \$1000 earns 6% annual interest for 5 years. Write the balance if it is compounded quarterly, then write the continuous-compounding balance. Use a calculator to approximate both amounts to the nearest cent and state which is larger.

Quarterly compounding has $n = 4$ periods per year, so

$$A_q = 1000 \left(1 + \frac{0.06}{4}\right)^{4(5)} = 1000(1.015)^{20} \approx \boxed{\$1346.86}.$$

For continuous compounding,

$$A_c = 1000e^{0.06(5)} = 1000e^{0.3} \approx \boxed{\$1349.86}.$$

The continuous-compounding balance is larger.

Exercise 10 (reasoning). Explain, using the formula $A(t) = P(1 + r/n)^{nt}$ and the definition of e , why continuous compounding gives a *larger* balance than compounding once a year, for the same P , r , and $t > 0$.

With annual compounding the balance is $P(1 + r)^t$; the continuous limit is $Pe^{rt} = P(e^r)^t$. For $r > 0$, the definition $e^r = \lim_{n \rightarrow \infty} (1 + r/n)^n$ gives $e^r > 1 + r$: for integer $n \geq 2$, the binomial expansion has first two terms $1 + r$, and its positive quadratic term is

$$\binom{n}{2} \left(\frac{r}{n}\right)^2 = \frac{r^2(n-1)}{2n} \geq \frac{r^2}{4} > 0.$$

Thus $e^r > 1 + r$. Raising these positive bases to $t > 0$ preserves their ordering, and multiplying by $P > 0$ shows $Pe^{rt} > P(1 + r)^t$. Therefore continuous compounding gives the larger balance.

Exercise 11 (features). For $y = 3 \cdot 2^x - 4$: state the horizontal asymptote, the y -intercept, and whether the graph is increasing or decreasing.

The graph is a vertical scale of 2^x by 3, followed by a shift down 4. Its horizontal asymptote is $y = -4$. At $x = 0$, $y = 3(1) - 4 = -1$, so the y -intercept is $(0, -1)$. Since 2^x is increasing and the scale factor is positive, the function is increasing.

Exercise 12 (transformations). For $y = -2^{x+1} + 3$: describe the transformations from $y = 2^x$ in order, then state the horizontal asymptote and the point corresponding to $(0, 1)$ on the parent graph.

Starting at $y = 2^x$, shift left 1 to get 2^{x+1} , reflect across the x -axis to get -2^{x+1} , then shift up 3. The horizontal asymptote is $y = 3$. The parent point $(0, 1)$ moves to $(-1, 1)$, then $(-1, -1)$, and finally $(-1, 2)$.

Exercise 13 (sketch and explain). Sketch $y = e^{x-2}$ and $y = e^x - 2$ on separate axes. Explain in one sentence why these two graphs are different, even though both come from shifting $y = e^x$ by 2.

For $y = e^{x-2}$, shift $y = e^x$ right 2: its horizontal asymptote stays $y = 0$ and $(0, 1)$ moves to $(2, 1)$. For $y = e^x - 2$, shift the parent down 2: its asymptote becomes $y = -2$ and $(0, 1)$ moves to $(0, -1)$. They are different because one shift is horizontal and the other is vertical.

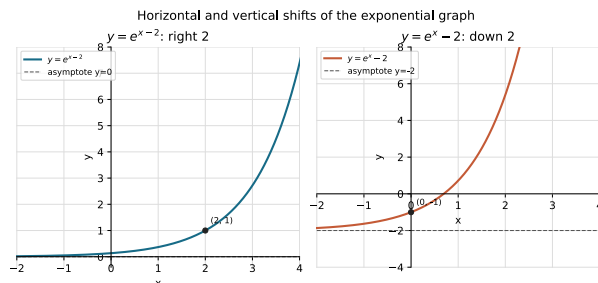


Figure 2: Separate plots of $y = e^{(x-2)}$ and $y = e^x - 2$, illustrating a horizontal shift versus a vertical shift

Exercise 14 (harder). A bacteria colony is modeled by $N(t) = 500 \cdot 2^{t/3}$, where t is measured in hours. Show directly that $N(t + 3) = 2N(t)$, then state the horizontal asymptote and explain what it means in the context of the model. (A later logarithms lecture will show how to rewrite this model using e .)

Substitute $t + 3$ and use the addition rule:

$$N(t + 3) = 500 \cdot 2^{(t+3)/3} = 500 \cdot 2^{t/3+1} = 2 (500 \cdot 2^{t/3}) = \boxed{2N(t)}.$$

The horizontal asymptote is $\boxed{N = 0}$. Mathematically, $500 \cdot 2^{t/3} \rightarrow 0$ as $t \rightarrow -\infty$, so the extended model approaches zero but never reaches it. In the colony context, time is ordinarily $t \geq 0$, where the model predicts growth from 500 bacteria; the asymptote describes the backward extension rather than a future extinction level.

Exercise 15 (calculator routine). A population of 5,000 bacteria grows 8% per hour. Write $P(t)$ in percentage-rate form and use a calculator to find the population after 3 hours, rounded to the nearest whole bacterium.

The per-hour growth multiplier is $1 + 0.08 = 1.08$, so

$$P(t) = \boxed{5000(1.08)^t}, \quad P(3) = 5000(1.08)^3 = 5000(1.259712) = 6298.56 \approx \boxed{6299 \text{ bacteria}}.$$

Exercise 16 (routine). A machine worth \$8,000 depreciates 20% per year. Write $V(t)$ in percentage-rate form, state whether this is growth or decay and why, and find its value after 3 years.

After losing 20%, the machine retains 80% of its value each year. Thus $V(t) = \boxed{8000(0.80)^t}$. This is decay because the multiplier 0.80 is between 0 and 1. After three years,

$$V(3) = 8000(0.80)^3 = 8000(0.512) = \boxed{\$4096}.$$

Exercise 17 (half-life). A radioactive sample has a half-life of 6 hours and starts at $A_0 = 80$ g. Find the amount remaining after 18 hours, without a calculator.

Eighteen hours is $18/6 = 3$ half-lives, so

$$A(18) = 80 \left(\frac{1}{2}\right)^{18/6} = 80 \left(\frac{1}{2}\right)^3 = 80 \cdot \frac{1}{8} = \boxed{10 \text{ g}}.$$

Exercise 18 (doubling time). A culture of yeast doubles every 2 hours, starting from 200 cells. Write $P(t)$ in doubling-time form, and find the population after 8 hours.

The doubling-time form is $P(t) = \boxed{200 \cdot 2^{t/2}}$. After 8 hours there have been $8/2 = 4$ doublings:

$$P(8) = 200 \cdot 2^4 = \boxed{3200 \text{ cells}}.$$

Exercise 19 (model from data). A culture has 300 cells at $t = 0$ and 1200 cells at $t = 4$ hours. Assume $P(t) = ab^t$.

- Find a and b .
- Write the model $P(t)$.
- Use it to predict the population after 6 hours.

At $t = 0$, $300 = P(0) = ab^0 = a$, so $a = 300$. At $t = 4$,

$$1200 = 300b^4 \implies b^4 = 4.$$

An exponential base is positive, so $b = \sqrt[4]{4} = \sqrt{2}$. Thus

$$\boxed{P(t) = 300(\sqrt{2})^t}.$$

After six hours,

$$P(6) = 300(\sqrt{2})^6 = 300 \cdot 8 = \boxed{2400 \text{ cells}}.$$

Exercise 20 (challenge). Show that the percentage-rate form $P(t) = P_0(1+r)^t$ and the doubling-time form $P(t) = P_0 \cdot 2^{t/d}$ describe the same function exactly when $1+r = 2^{1/d}$ – that is, a growth rate r per period corresponds to doubling time d .

For the two formulas to agree for every t , their values at one period, $t = 1$, must agree. Assuming $P_0 > 0$, divide both by P_0 to get

$$(1+r)^1 = 2^{1/d}.$$

This condition is necessary. Conversely, if $1+r = 2^{1/d}$, then for every t ,

$$P_0(1+r)^t = P_0 \left(2^{1/d}\right)^t = P_0 \cdot 2^{t/d},$$

so the formulas agree exactly. The condition is therefore $\boxed{1+r = 2^{1/d}}$.